

Mariam Ali Akbar

Software Engineer | Full-Stack Developer

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SUMMARY

Computer Science graduate (CGPA 3.7/4.0) building full-stack web applications with React, Next.js, TypeScript, Node.js and PostgreSQL. Hands-on experience in REST API development, AI/RAG systems, HubSpot automation and CRM data workflows through internships at Fantech Labs and the Punjab Information Technology Board (PITB).

EXPERIENCE

Software Engineering Intern | Fantech Labs Jul 2026 – Sep 2026

- Built a **Movie Explorer** application using Next.js and TypeScript with dynamic movie pages, debounced search and favorites.
- Built and reviewed a forms project using React Hook Form, Zod, Prisma and PostgreSQL.
- Built an API to create HubSpot workflows programmatically via Claude Code, covering a gap where the HubSpot MCP server did not support workflow creation.
- Worked with HubSpot workflows, data enrichment, CRM integrations, orphan company/contact identification, and Metadata.io forms and metadata.
- Used Google Tag Manager and Google Analytics, and reviewed Metadata.io tags for digital marketing and analytics tasks.

Backend Intern (Hybrid) | Punjab Information Technology Board (PITB), Lahore Apr 2026 – Jun 2026

- Developed backend APIs using Express.js for CRUD operations, routing, file uploads and JWT-based authentication.
- Tested REST APIs with Postman and integrated relational data using MySQL.

Frontend Intern | Bytewise Limited 2024

- Developed responsive web interfaces using HTML, Tailwind CSS and JavaScript.
- Built interactive React.js projects including an Expense Tracker, Shopping Cart and To-Do application.
- Practiced component-based architecture, Git version control and Vercel deployment.

PROJECTS

Movie Explorer | Next.js, TypeScript, TMDB API, Tailwind CSS, shadcn/ui Internship Project

- Built a movie discovery application with trending, popular, top-rated, upcoming and now-playing sections, dynamic movie pages, search, favorites and recently viewed movies.
- Implemented debounced movie search with URL-based search state and keyboard-accessible search interactions.
- Built reusable components with shadcn/ui and deployed the application on Vercel.

WordKnit | React, Node.js, Express.js, MongoDB, Python, Docker, RAG, LLM APIs Final Year Project

- Built a full-stack vocabulary and reading companion with AI-powered contextual word explanations and speaking/vocabulary coaching features.
- Implemented a RAG pipeline using TF-IDF vectorization and PyMuPDF to retrieve contextual definitions from vocabulary resources.
- Integrated an LLM to generate contextual word explanations, synonym nuances and memory-oriented learning support.
- Built document/PDF processing features using PyMuPDF and PDF.js.

RAG-Based Document Chatbot | Next.js, Node.js, PostgreSQL, Vector Search, LLM Project

- Built a chatbot that answers questions about uploaded documents using retrieval-augmented generation (RAG).
- Split documents into chunks and used embeddings and semantic search to find the most relevant passages for each question.
- Showed the source passages behind each answer so users can check it against their documents.

E-Commerce Application | React.js, JavaScript Personal Project

- Built an e-commerce application to strengthen understanding of React components, state management and React hooks.

EDUCATION

BS Computer Science | University of Okara, Renala Khurd 2022 – 2026

CGPA: 3.7/4.0

CERTIFICATIONS & COURSEWORK

- CS50x — Harvard / edX
- Coursework: Data Structures & Algorithms, Database Systems, PostgreSQL, React, Next.js, TypeScript

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, SQL

Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS, shadcn/ui

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Databases: PostgreSQL, MongoDB, MySQL, Prisma

AI: RAG, LLM Integration, TF-IDF

Tools & Platforms: Git, GitHub, Docker, Vercel, Postman, HubSpot, Google Tag Manager, Google Analytics